Technology Stack

Date	26 June 2026
Team ID	SWTID-2026-5060
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that will be used to build the Credit Card Approval Prediction System. A well-chosen technology stack ensures that the application is accurate, scalable, maintainable, and capable of providing fast and reliable credit card approval predictions.

Technology Stack Details

S.NO	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, JavaScript, Jinja2 Templates	HTML, CSS, JavaScript, and Jinja2 templates were used to develop a responsive and interactive web interface for collecting applicant information and displaying prediction results along with detailed risk audit reports.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java	Python, Flask	Flask was selected because it is lightweight, easy to integrate with machine learning models, and efficiently handles routing, request processing, and prediction execution
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	JSON Files (prediction_summary.json), Joblib Model	JSON files are used to store prediction history and audit logs, while the trained Random Forest model is stored as a Joblib file for efficient loading and prediction.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Docker, Render	Docker provides a consistent deployment environment through containerization, while Render enables simple and reliable cloud deployment of the Flask application.

5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub	Git and GitHub are used for source code management, version tracking, collaboration, and maintaining project history throughout the development lifecycle.
6	Third-Party APIs / Other Tools (e.g. Stripe, Twilio, Google Maps)	Scikit-learn, Pandas, NumPy, Joblib	Scikit-learn is used to train and deploy the Random Forest model, Pandas and NumPy perform data preprocessing and feature engineering, and Joblib is used for model serialization and loading.